Technical Document #271: Software Engineering - Comprehensive Overview

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Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Key Takeaways:

- Refactoring improves code structure without changing behavior.
- Continuous deployment (CD) automatically deploys code changes to production.
- Microservices architecture structures applications as independently deployable services.